

User Guide
for
Formula Subset Analysis (FSA)

Author: Ke-Shiuan Lynn Ph.D.

Assistant Professor
Department of Mathematics
Fu-Jen Catholic University
Email: 128171@mail.fju.edu.tw

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Noted that the datasets excerpted from PubChem and HMDB were downloaded in 2018 and thus the contents in the current web pages now may be different from the snapshots below. Also, the file names and the corresponding path may need to be changed in the provided programs. In addition, the provided Python programs are better run under python 3.6 or later and the MATLAB programs are better run under MATLAB 2021b or later.

1. Download Data:

- a 、 PubChem: For the search of compound mass formula candidates

(<https://ftp.ncbi.nlm.nih.gov/pubchem/Compound/CURRENT-Full/XML/>)



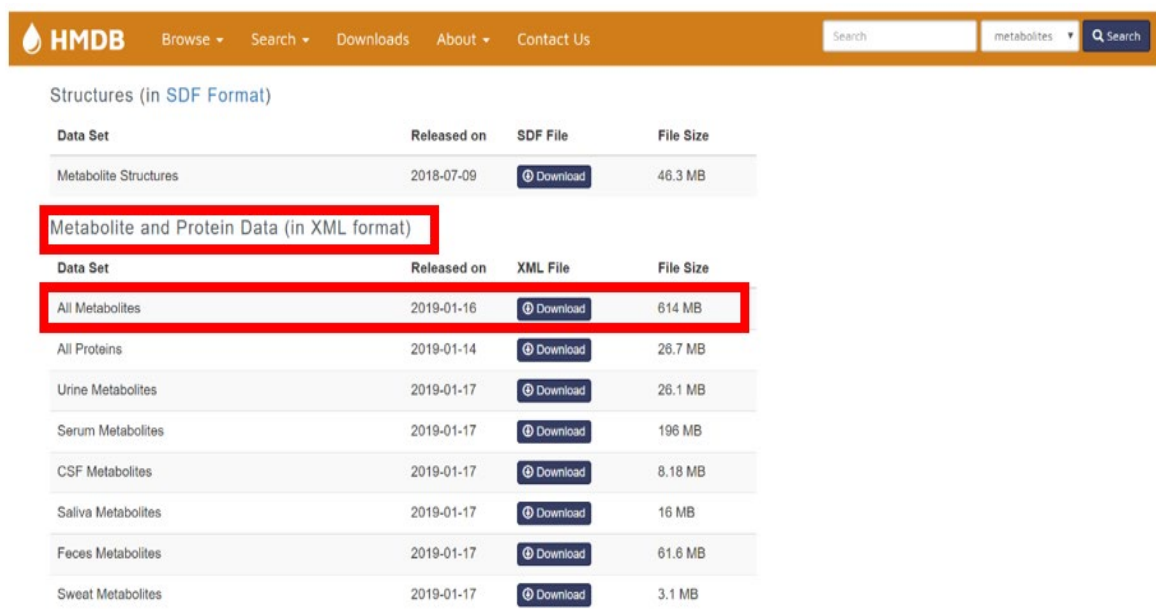
/pubchem/Compound/CURRENT-Full/XML

[父目録]

名	大	已
稱	小	修
		改
		日
		期
Compound_000000001_000025000.xml.gz	41.4 MB	2019/2/7 上午2:07:00
Compound_000025001_000050000.xml.gz	46.1 MB	2019/2/7 上午2:07:00
Compound_000050001_000075000.xml.gz	46.2 MB	2019/2/7 上午2:07:00
Compound_000075001_000100000.xml.gz	38.9 MB	2019/2/7 上午2:06:00
Compound_000100001_000125000.xml.gz	49.4 MB	2019/2/7 上午2:07:00
Compound_000125001_000150000.xml.gz	42.8 MB	2019/2/7 上午2:07:00
Compound_000150001_000175000.xml.gz	51.5 MB	2019/2/7 上午2:07:00
Compound_000175001_000200000.xml.gz	52.3 MB	2019/2/7 上午2:08:00
Compound_000200001_000225000.xml.gz	50.0 MB	2019/2/7 上午2:06:00
Compound_000225001_000250000.xml.gz	44.6 MB	2019/2/7 上午2:08:00
Compound_000250001_000275000.xml.gz	46.6 MB	2019/2/7 上午2:07:00
Compound_000275001_000300000.xml.gz	48.2 MB	2019/2/7 上午2:08:00
Compound_000300001_000325000.xml.gz	46.7 MB	2019/2/7 上午2:09:00
Compound_000325001_000350000.xml.gz	48.5 MB	2019/2/7 上午2:08:00
Compound_000350001_000375000.xml.gz	51.5 MB	2019/2/7 上午2:09:00

- b 、 HMDB Test dataset (<http://www.hmdb.ca/downloads>)

i. Compound information



Data Set	Released on	SDF File	File Size
Metabolite Structures	2018-07-09	Download	46.3 MB

Data Set	Released on	XML File	File Size
All Metabolites	2019-01-16	Download	614 MB
All Proteins	2019-01-14	Download	26.7 MB
Urine Metabolites	2019-01-17	Download	26.1 MB
Serum Metabolites	2019-01-17	Download	196 MB
CSF Metabolites	2019-01-17	Download	8.18 MB
Saliva Metabolites	2019-01-17	Download	16 MB
Feces Metabolites	2019-01-17	Download	61.6 MB
Sweat Metabolites	2019-01-17	Download	3.1 MB

ii. MS/MS spectra

HMDB Browse Search Downloads About Contact Us			
Saliva Metabolites	2019-01-17	Download	16 MB
Feces Metabolites	2019-01-17	Download	61.6 MB
Sweat Metabolites	2019-01-17	Download	3.1 MB
Spectra			
Data Set	Released on	Download Link	File Size
Mass Spectra Image Files	2019-02-01	Download	166 MB
NMR Spectra FID Files	2019-02-01	Download	1.92 GB
Raw NMR Spectra Peaklist Files (TXT)	2019-02-01	Download	918 KB
Raw GC-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	23 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Predicted	2019-02-01	Download	158 MB
Raw MS-MS Spectra Peaklist Files (TXT) - Experimental	2019-02-01	Download	2.28 MB
All Raw Spectra Peaklist Files (TXT)	2019-02-01	Download	1.68 GB
NMR Spectra Files (XML)	2019-02-01	Download	4.46 MB
GC-MS Spectra Files (XML) - Predicted	2019-02-01	Download	69 MB
GC-MS Spectra Files (XML) - Experimental	2019-02-01	Download	17.1 MB
MS-MS Spectra Files (XML) - Predicted	2019-02-01	Download	485 MB
MS-MS Spectra Files (XML) - Experimental	2019-02-01	Download	38.1 MB
All Spectra Files (XML)	2019-02-01	Download	4.1 GB

c、GNPS NIH Natural Products Library:

Go to the website at <https://gnps-external.ucsd.edu/gnpslibrary>, find the rows shown in the following figure, and click on the "Download" links under the "MGF Download" column.

GNPS Library List

GNPS Library	Library Type	MGF Download	MSP Download	JSON Download
GNPS-LIBRARY	GNPS	Download	Download	Download
GNPS-SELLECKCHEM-FDA-PART1	GNPS	Download	Download	Download
GNPS-SELLECKCHEM-FDA-PART2	GNPS	Download	Download	Download
GNPS-PRESTWICKPHYTOCHEM	GNPS	Download	Download	Download
GNPS-NIH-CLINICALCOLLECTION1	GNPS	Download	Download	Download
GNPS-NIH-CLINICALCOLLECTION2	GNPS	Download	Download	Download
GNPS-NIH-NATURALPRODUCTSLIBRARY	GNPS	Download	Download	Download
GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_POSITIVE	GNPS	Download	Download	Download
GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_NEGATIVE	GNPS	Download	Download	Download
GNPS-NIH-SMALLMOLECULEPHARMACOLOGICALLYACTIVE	GNPS	Download	Download	Download

d、GNPS Pacific Northwest National Lab Lipids Library

Go to the website at <https://gnps-external.ucsd.edu/gnpslibrary>, find the rows shown in the following figure and click on the "Download" links under the "MGF Download" column.

DEREPlicATOR_IDENTIFIED_LIBRARY	GNPS	Download	Download	Download
PNNL-LIPIDS-POSITIVE	GNPS	Download	Download	Download
PNNL-LIPIDS-NEGATIVE	GNPS	Download	Download	Download
MIADB	GNPS	Download	Download	Download

2. Use **MySQL Workbench** to construct a **compound** database
 - a 、 Install **MySQL Workbench** and create a user "root" with password "1234" (MySQL Workbench Installation: A Step-by-Step Guide <https://www.simplilearn.com/tutorials/mysql-tutorial/mysql-workbench-installation>).
 - b 、 Add a "**compound**" database
 - i. Right-click on the list of existing **Schemas** and select **Create Schema**.
 - ii. Enter "**compound**" for the schema and for collation choose '**utf - utf8_bin**'. Then click **Apply**.
 - iii. In the **Apply SQL Script to Database** window, click **Apply** to run the SQL command that creates the schema.
 - iv. Click **Finish**. You can see the new schema, which has no tables, listed in the left pane.
 - v. From the **Management** menu, select **Users and Privileges** and click **Add Account**. Complete the screen with the credentials listed above. Navigate to the tab **Schema Privileges** and click on **Add Entry....** Select the newly created database schema. Click **Select "All"** to grant all privileges on this schema for this new user. Click **Apply**.
 - vi. From the **Management** menu, select **Options File** and click the **Networking** tab. Find the **max_allowed_packet** entry (should be at the top) and change it to at least **256M**. Click **Apply...** and in the new dialog that appears again **Apply**.
 - vii. If you get the error *Could not save configuration file*, you did not run the MySQL Workbench as an administrator. Restart it as an administrator and try again.
 - viii. Finally, to apply this change you need to restart the database. From the **Instance** menu, select **Startup/Shutdown** and click **Stop Server**, followed by **Start Server**.
3. Construct PubChem Dataset for Compound Formula Candidates Search
 - a 、 Use the provided [readgzip_COMPOUNDtoMYSQL.py](#) to convert PubChem XML files into MySQL database table "**compounddata20190529**".
 - b 、 Be sure to adjust the user name and password of the SQL server and the file names and paths of the downloaded files if different names are used.
 - c 、 Use the following query to extract the qualified records and save them into "[PubChem_metabolites.csv](#)"


```
SELECT MolecularFormula, Weight_MonoIsotopic
FROM compounddata20190529 as t1
WHERE t1.Weight_MonoIsotopic < 1500 and (t1.MolecularFormula like '%C%' or
t1.MolecularFormula like '%H%' or t1.MolecularFormula like '%O%' or
t1.MolecularFormula like '%N%' or t1.MolecularFormula like '%S%' or
t1.MolecularFormula like '%P%') and t1.MolecularFormula not like '%F%' and
t1.MolecularFormula not like '%Cl%' and t1.MolecularFormula not like '%Br%' and
t1.MolecularFormula not like '%I%'
INTO OUTFILE 'your_SQL_directory/Uploads/PubChem_metabolites.csv'
```

FIELDS TERMINATED BY ','

LINES TERMINATED BY '\r\n';

- d 、 Use Matlab program "[convert_PubChem_metabolites.m](#)" to further ensure the element contains CHONSP only and to convert the formula into a numerical matrix indicating number of CHONSP elements in each formula. The "[PubChem_metabolites.csv](#)" is converted into output file "[PubChemMetabolite.mat](#)" which is need in all the FSA tests in step 5.

4. Construct test datasets

a 、 HMDB

- i. Use the provided python file [HMDBtoMySQL4_0218.py](#) to convert HMDB Compound XML file into MySQL database table "[hmdbdata20190226](#)".
- ii. Use the provided python files [MSMStoMySQL.py](#) and [MSMStoMySQL_otherproblem.py](#) to convert HMDB MS/MS XML file into MySQL database table "[msmsdata20190226](#)"
- iii. Use the follow query to extract the qualified records

```
select t1.name, t1.accession, t1.chemical_formula, t1.monisotopic_molecular_weight,  
      t2.ionization_mode, t2.instrument_type, t2.collision_energy_voltage, t2.peak_counter,  
      t2.ms2_peak  
from hmdbdata20190226 as t1  
inner join msmsdata20190226 as t2  
on t2.database_id=t1.accession  
where (t2.instrument_type like '%LC-ESI-QTOF%' or t2.instrument_type like '%LC-ESI-  
ITFT%' or t2.instrument_type like '%LC-ESI-QFT%') and t2.peak_counter > 2 and  
t1.chemical_formula like '%C%' and t1.chemical_formula like '%H%' and  
t1.chemical_formula not like '%F%' and t1.chemical_formula not like '%Cl%' and  
t1.chemical_formula not like '%Br%' and t1.chemical_formula not like '%I%' and  
t1.chemical_formula not like '%B%';
```
- iv. Use the "Export" function in MySQL Workbench to save the resultant record to an xml file "[HMDB_CHONPS_MSMS.xlsx.xml](#)"
- v. Open the file with MS Excel and save it as "[HMDB_CHONPS_MSMS.xlsx](#)".

b 、 GNPS NIH Natural Products Library:

- i. Put the provided 3 Matlab files [GNPS_NPL_mgf2mat.m](#), [readNatProdMGF.m](#), and [readGNPS_NatProdR2_MGF.m](#) along with the 3 data files [GNPS-NIH-NATURALPRODUCTSLIBRARY.mgf](#), [GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_NEGATIVE.mgf](#), and [GNPS-NIH-NATURALPRODUCTSLIBRARY_ROUND2_POSITIVE.mgf](#) downloaded in step 1(c) into the same directory.
- ii. Run [GNPS_NPL_mgf2mat.m](#) in Matlab (2021a or a later version) and you should have 3 Matlab data files [GNPS_NPL_MAXIS.mat](#), [GNPS_NPL_QTOF.mat](#), and

[GNPS_NPL_ORBITRAP.mat](#) in the directory.

c 、 GNPS Pacific Northwest National Lab Lipids Library:

- i. Put the provided 3 Matlab files [GNPS_PNNL_LIPIDS_mgf2mat.m](#) and [readGNPS_PNNL_LIPIDS_MGF.m](#) along with the 2 data files [PNNL-LIPIDS-NEGATIVE.mgf](#) and [PNNL-LIPIDS-POSITIVE.mgf](#) downloaded in step 1(d) into the same directory.
- ii. Run [GNPS_PNNL_LIPIDS_mgf2mat.m](#) in Matlab (2021a or a later version) and you should have the Matlab data file [PNNLLIPID_spectra.mat](#) in the directory.

5. Run FSA on the test datasets in Matlab (2021a or a later version)

- a 、 Use the provided [FSA_all7datasets.m](#) and put the 8 previous generated [PubChemMetabolite.mat](#), [HMDB_QTOF_spectra.mat](#), [HMDB_QTOF_spectra.mat](#), [HMDB_QTOF_spectra.mat](#), [GNPS_NPL_MAXIS.mat](#), [GNPS_NPL_QTOF.mat](#), [GNPS_NPL_ORBITRAP.mat](#), and [PNNLLIPID_spectra.mat](#) in the same directory. Users can download the provided [FSA_HMDB_result_CHONSP.xlsx](#) and [PubChemMetabolite.mat](#) since they are authorized to be distributed here.
- b 、 Run [FSA_all7datasets.m](#) and you should have the output file [Supplementary_File1.xlsx](#), which includes the formula ranking results of the 7 datasets.

6. Run FSA on user's dataset in Matlab (2021a or later version)

- a 、 A single peak list from a MS/MS spectrum:
 - i. Modify lines 14~19 of the provided [SingleSpectrumFSA.m](#) in Matlab
Line14: The precursor m/z value
Line15: The m/z values of the peaks in a centroided MS/MS spectrum
Line16: The corresponding abundances of the peaks (in percentage) in line 15
Line17: Mass matching tolerance (can be either in Da (if value < 1) or in PPM (if value ≥ 1))
Line18: Max number of ranked formulas for each spectrum

```
14 precursor_mz=245.2341;  
15 mz=[245.2341, 171.1502, 129.1395, 112.1128, 100.0760, 84.0812];  
16 ab=[56.8106, 58.1811, 65.6561, 90.0748, 100.0000, 51.4535];  
17 mode='positive';  
18 Match_Tol=0.001;  
19 MaxQualifiedResult=5;
```

ii. Run the modified program in Matlab

b 、 Multiple peak lists from a MS/MS spectral data set:

- i. Generate a MS Excel file of the peak lists. Note that in a peak list, m/z and abundance (or intensity) values should appear in pairs and all values should be separated by a semicolon (“;”). An example file “[MSMS_peaklist_example.xlsx](#)” is provided.
- ii. Modify lines 19~23 of the provided “[MultipleSpectraFSA.m](#)”.
Line19: Threshold (in ratio) to remove small peaks
Line20: mass matching tolerance

Line21: Max number of ranked formulas for each spectrum

Line22: File name of the output formula ranking result.

Line23: File name of the peak lists as generated in step i.

```
19 opt.TINY=0; % ratio threshold for "tiny" peaks
20 opt.PPM=0.001;% mass match tolerance
21 opt.MaxRankNumber=5;
22 output_filename='Precursor_Formula_Ranking_Result_CHONSP.xlsx'; % file name of the formula ranking result
23 MS2Table=readtable('MSMS_peaklist_example.xlsx','FileType','spreadsheet',...
24 'TextType','string','VariableNamingRule','modify'); % file name of the spectral peak list
```

iii. Run the modified program in Matlab